DustiniaDelixia_Groceria Delivery Analytics: Optimizing E-Commerce Logistics Performance Using ClickHouse and Metabase

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***Abstract*—This paper presents a comprehensive business intelligence dashboard for analyzing delivery performance in e-commerce operations. Using the DustiniaDelixia_Groceria dataset provided by Lab MCI, comprising 96,984 orders, we developed an analytical system leveraging ClickHouse as an OLAP database and Metabase for visualization. The analysis reveals an on-time delivery rate of 91.89% with an average delivery time of 12.56 days. Critical findings include: (1) carrier transit accounts for 74% of total delivery time (9.33 days), identifying it as the primary bottleneck; (2) specific regional states exhibit 2-3× higher delay rates compared to the national average of 8.11%; (3) delayed orders cause a 1.72-point drop in customer review scores (4.29 vs 2.57). The dashboard provides actionable insights for operational optimization, including seller performance monitoring, regional bottleneck identification, and customer satisfaction impact analysis.**

***Index Terms*—business intelligence, delivery optimization, e-commerce logistics, ClickHouse, Metabase, data analytics**

I. Introduction

The rapid growth of e-commerce has intensified competition in delivery performance, making on-time delivery a critical differentiator for customer satisfaction and retention. E-commerce operations face unique logistical challenges due to geographic dispersion and infrastructure disparities across regions.

This research addresses three key questions: (1) What are the primary bottlenecks in the delivery pipeline? (2) How do regional disparities affect delivery performance? (3) What is the quantifiable impact of delays on customer satisfaction?

We developed an end-to-end analytics dashboard that transforms raw e-commerce data into actionable operational insights. The system processes 96,984 orders from the DustiniaDelixia_Groceria dataset using a modern data stack comprising ClickHouse for high-performance analytics and Metabase for interactive visualization.

II. Methodology

A. System Architecture

The data pipeline follows a star schema design pattern. Raw CSV files are ingested into ClickHouse, processed into a fact table (`fact_operational_delivery`) and dimension tables, and finally visualized via Metabase.

As shown in Fig. 1, the architecture ensures sub-second query performance on large datasets while providing interactive filtering capabilities for end-users.

B. Key Metrics Calculation

The following metrics are computed using advanced SQL queries in ClickHouse:

$$\text{Delay Rate} = \frac{\text{Delayed Orders}}{\text{Total Orders}} \times 100 \tag{1}$$

$$\text{On-Time Rate} = \frac{\text{Total Orders} - \text{Delayed Orders}}{\text{Total Orders}} \times 100 \tag{2}$$

Delivery stages are broken down into three phases: Approval Time, Seller Processing, and Carrier Transit.

III. Results and Discussion

A. Overall Performance Metrics

The analysis of 96,984 orders reveals an On-Time Delivery Rate of **91.89%**, a Delay Rate of **8.11%**, and an Average Delivery Time of **12.56 days**. The average review score across all orders is **4.16/5**.

Fig. 2 illustrates the executive summary page, providing a high-level overview of the KPIs and monthly delay trends.

B. Bottleneck Analysis

Stage breakdown analysis identifies carrier transit as the critical bottleneck. Approval Time takes 0.43 days (3.4%), Seller Processing takes 2.82 days (22.5%), and Carrier Transit takes **9.33 days (74.1%)**.

As depicted in Fig. 3, logistics carrier performance, rather than seller operations, is the primary constraint in the delivery pipeline.

C. Regional and Customer Impact

Geographic analysis reveals severe performance disparities. Northeast states exhibit delay rates 2-3× higher than the national average of 8.11%, with Maranhão (MA) reaching 19.58%.

Furthermore, the correlation between delivery performance and customer satisfaction is severe. On-time orders receive a review score of **4.29**, while delayed orders drop to **2.57** (a

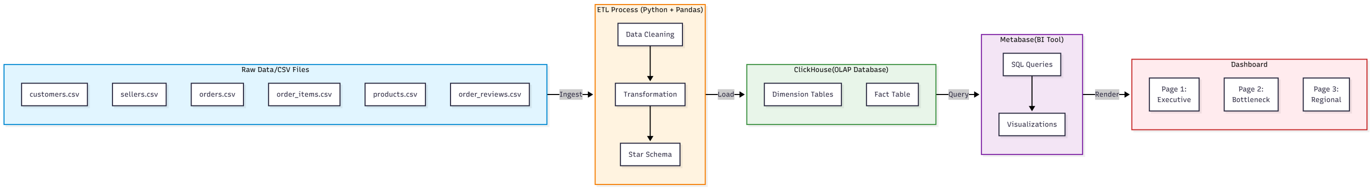

Fig. 1. Arsitektur Data Pipeline: DustiniaDelixia_Groceria Dashboard.

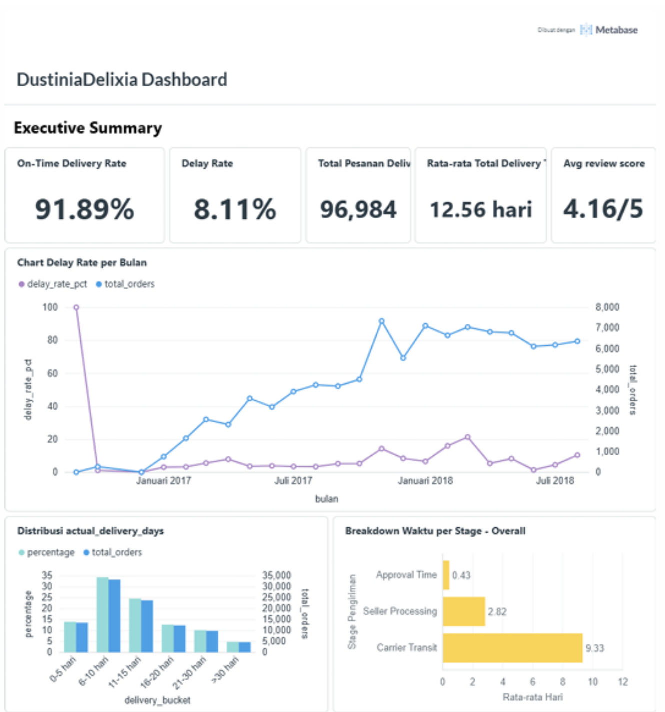

Fig. 2. Tampilan Dashboard Halaman 1: Executive Summary dan KPI Makro.

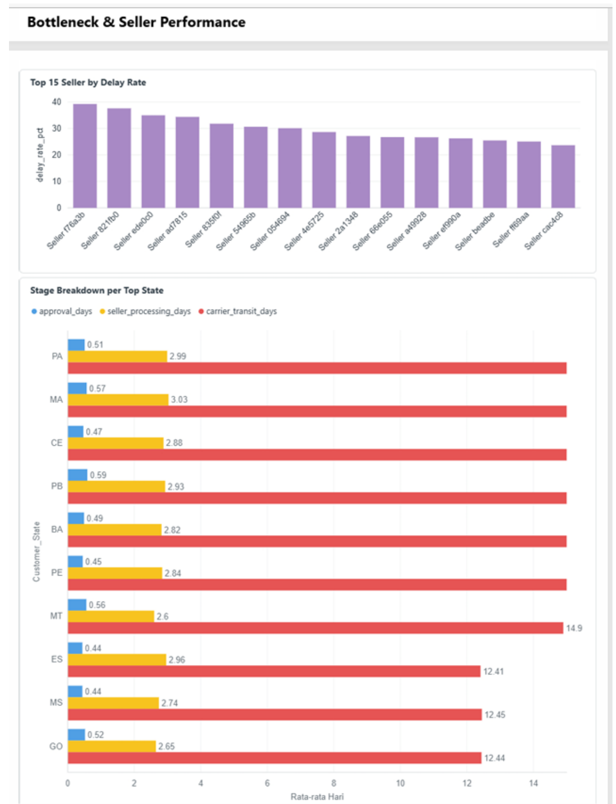

Fig. 3. Breakdown Waktu per Stage: Carrier Transit mendominasi 74% total waktu pengiriman.

1.72-point decrease). Orders delayed by more than 14 days receive a score of only 1.71.

Fig. 4 highlights the geographic hotspots and the direct negative impact of delays on customer trust.

IV. Business Recommendations

Based on the analytical findings, we propose the following actionable strategies:

- **Carrier Optimization:** Renegotiate contracts with carriers serving high-delay regional routes and establish regional distribution centers.
- **Seller Management:** Incentivize sellers who process orders within 24 hours and implement SLA penalties for chronic underperformers.
- **Customer Experience:** Launch a "First Order Guarantee" for new customers and set realistic delivery expectations (+1 day) for Friday-Saturday orders.

V. Conclusion

This research demonstrates the power of modern business intelligence tools in transforming operational data into strategic insights. The dashboard successfully identifies carrier transit as the primary bottleneck (74% of delivery time), quantifies severe regional disparities, and establishes the critical impact of delays on customer satisfaction. The implemented system using ClickHouse and Metabase provides real-time, actionable intelligence that enables data-driven decision-making for logistics optimization.

Acknowledgment

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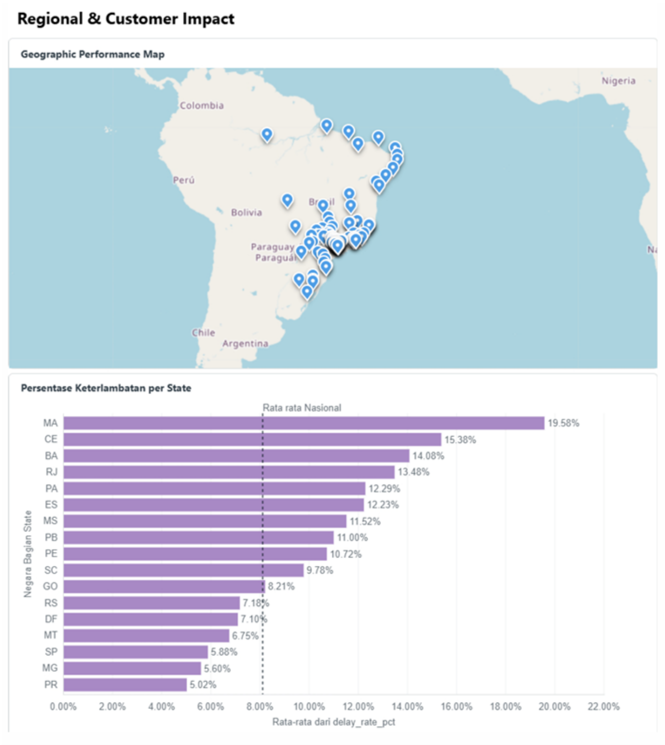

Fig. 4. Disparitas Regional dan Dampak Delay terhadap Review Score Pelanggan.

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